17 Oct 2025

AF Seminar 2

1. Noduri critice

2. Muchii critice

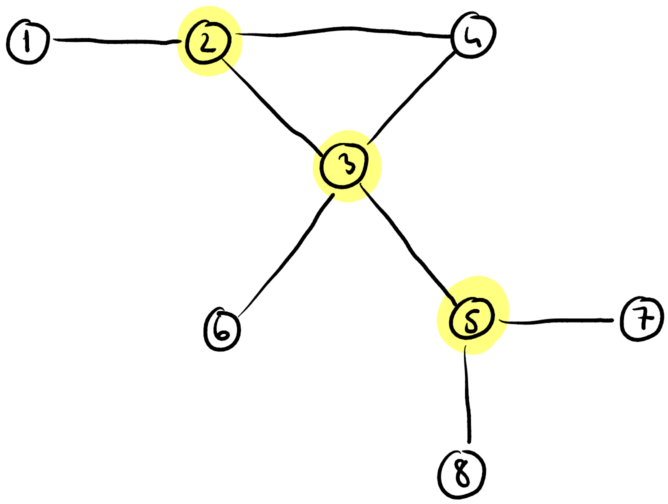
3. Algoritm

4. Proprietăți ale parcurgerilor

5. Grafuri bipartite

Noduri critice

- doar în G neorientate



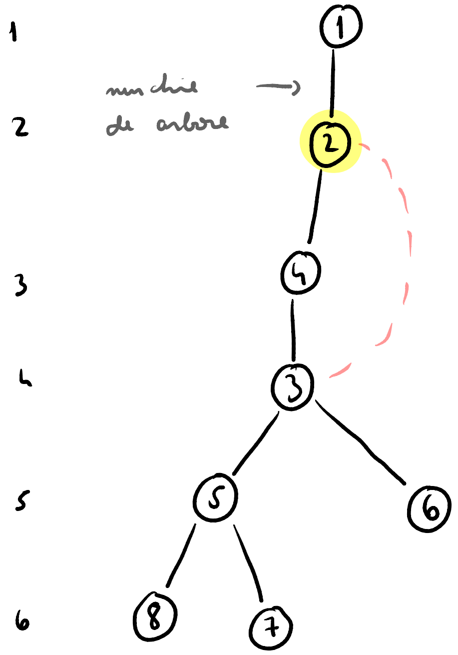
Brute force $O(n(n+m))$

Optim : $O(n+m)$

niv[x] = nivelul vf.-lui x în arborele DFS

low[x] = cel mai mic nivel la care pot ajunge din x printr-o muchie de întoarcere a lui sau a unui descendent / sau niv[x]

Arborele DFS



(orice muchie care nu apare în arbore => m. întoarcere)

	1	2	3	4	5	6	7	8
niv	1	2	4	3	5	5	6	6

	1	2	3	4	5	6	7	8
low	1	2	2	2	5	5	6	6

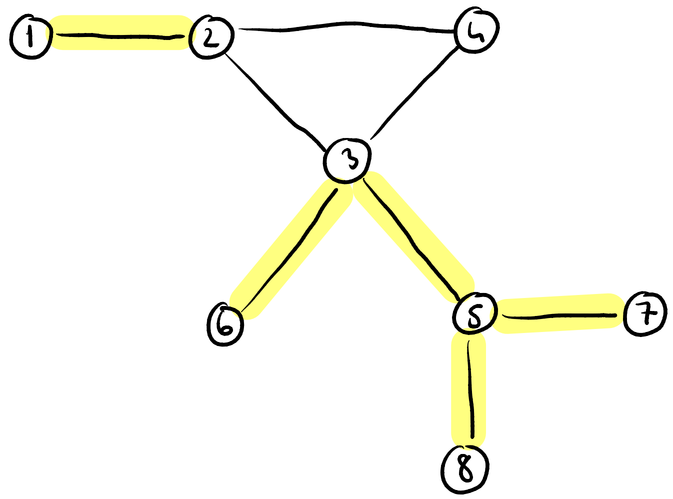
x - nod critic <=> $\exists$ y $\in$ fiu[x] a.î. low[y] $\geq$ niv[x]

x - nu e rădăcină

Dacă x rad atunci

x nod critic <=> x cu cel puțin 2 fii

Muchii critice



muchiile de întoarcere nu pot fi m. critice

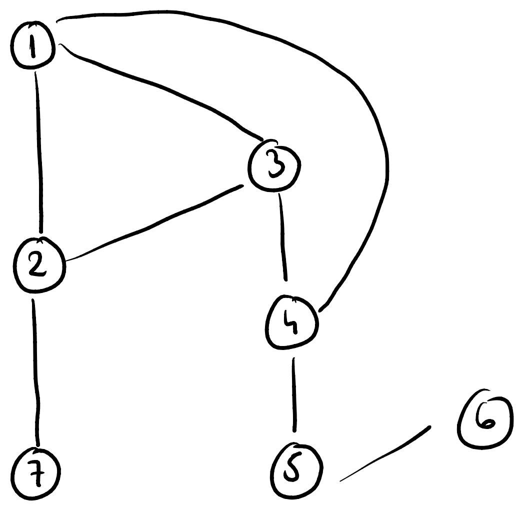
(x, y) muchie critică $\Leftrightarrow$ $low[y] > niv[x]$

Notăm

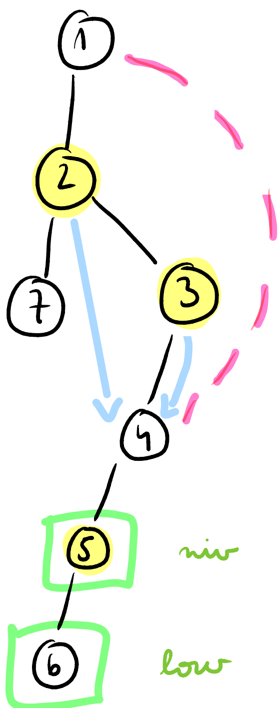
- vectorul niv
- vectorul low
- G [x] - lista de adiacență a lui x

Algoritm

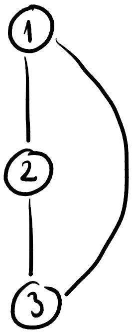
```
                                  părintele
                          nodul curent
void DFS ( int nod , int p )

    niv [ nod ] = niv [ p ] + 1
    low [ nod ] = niv [ nod ]

    for ( auto vecin : G [ nod ] )
    {
        if ( niv [ vecin ] == 0 )
        {
            DFS ( vecin , nod ) ;
            low [ nod ] = min ( low [ nod ], low [ vecin ] )
            if ( low [ vecin ] >= niv [ nod ] )
                crit [ nod ] = 1
        }
        else
            low [ nod ] = min ( low [ nod ],
                                niv [ vecin ]);
    }
```



	1	2	3	4	5	6	7
niv	1	2	3	4	5	6	3
low	1	2	~~3~~ 1	~~4~~ 1	5	6	3



exemple 2

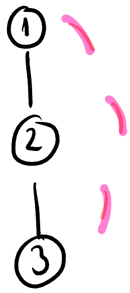


low

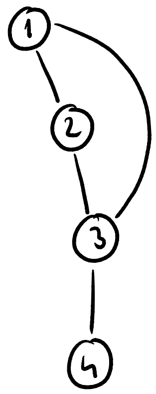
	1	
1	~~2~~	~~3~~
		1

min

1	2	3



Proprietăți ale parcurgerilor



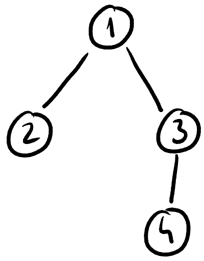
d :

1	2	3	4
0	1	1	2

BFS

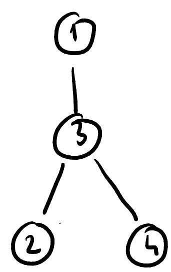
Queue : ~~1~~, ~~2~~, ~~3~~, 4

coadă



Arborele BFS

DFS



Arbore DFS

arborele DFS = arborele BFS $\Leftrightarrow$ G arbore

ex 1

Nr. maxim de muchii pe care le putem el. ca vectorul de distanțe să rămână același

n noduri

m muchii

$$m - (n-1)$$

ex 2

Fie $G_1 = (V_1, E_1)$
$G_2 = (V_2, E_2)$ conexe

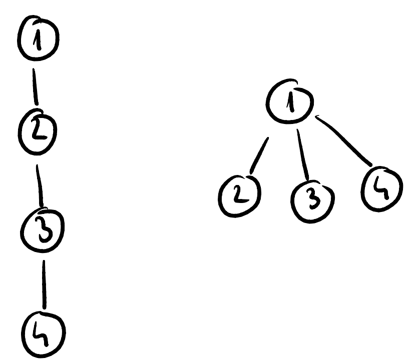
$|E_1| = |E_2|$ (au același nr. de muchii)

$DFS_{G_1} = DFS_{G_2}$

$BFS_{G_1} = BFS_{G_2}$

+

$\Rightarrow$ este $G_1 = G_2$?



\# NU

□

ex 3

Det unui ciclu într-un graf orientat

#

Facem DFS => muchie de întoarcere => ciclu

$\square$

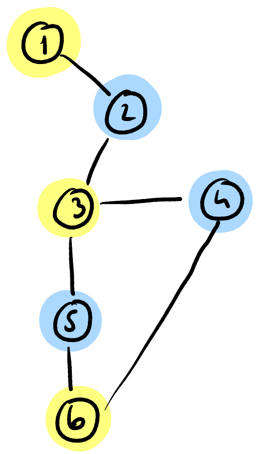
ex 4

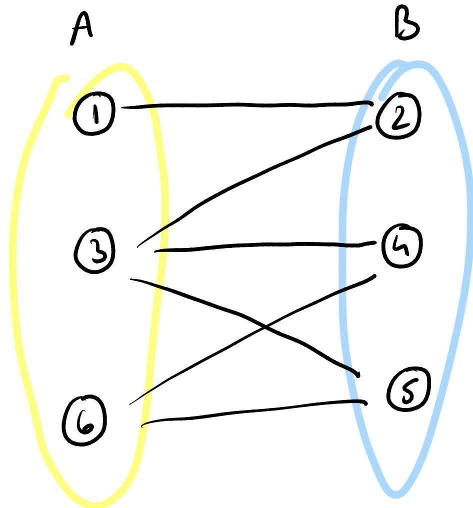
Fie un G cu n noduri și m muchii

condiția ca G să conțină un ciclu ?

$$m > n - 1$$

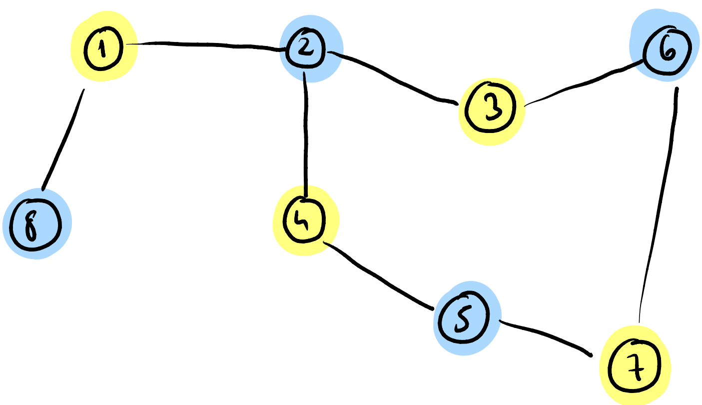
Grafuri bipartite

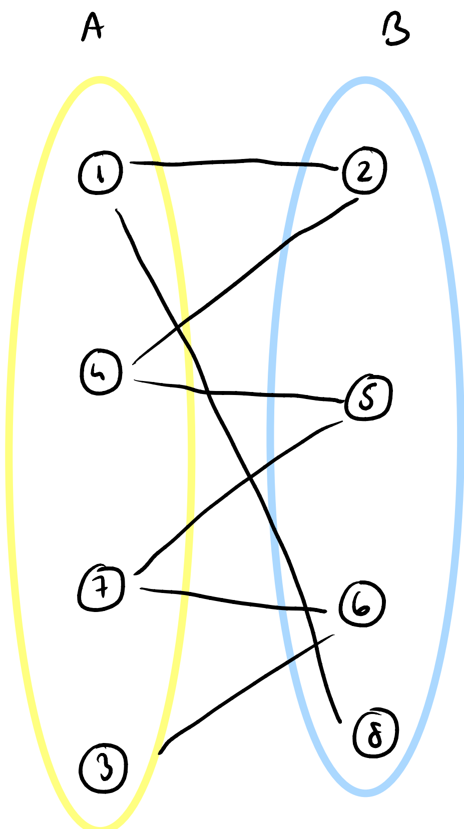




G bipartit $\iff$ G nu are ciclu de lungime impară

$\iff$ 2 colorabil





```
culoare [1] , DFS(1)

void DFS ( int nod )

    for ( auto vecin : G[ nod ] )

        if ( cul [ vecin ] == 0 )
        {
            cul [ vecin ] = 3 - c [ nod ]       daca culoarea
                                                  = 1 => 2
            DFS ( vecin )                         = 2 => 1
        }
        else {
            if ( cul [ vecin ] == cul [ nod ] )
                ok = False
        }
```